Chair of Product Development and Lightweight Design
Technical University of Munich

Lab Course Report

"Scientific Experiment with Research Data Management – Tensile Test"

Prof. Dr. Markus Zimmermann
Winter Semester 2026/2027

Group: 1	
Last name:	**First name:**
Müller	Georg
Qiu	Junjie
Radlach	Mark
Streicher	Martin

Test date: 13.10.2026 Submission: 27.10.2026

English translation of the report originally submitted in German.

Contents

1 Description of the Test Objects **2**
- 1.1 Working Principle of the Testing Machine . . . 2
- 1.2 Control, Drive and Clamping . . . 2
- 1.3 Sources of Error . . . 3
- 1.4 Force Measurement . . . 4
- 1.5 Strain Measurement Methods – Transducers . . . 4
- 1.6 Plotting . . . 5

2 Interpretation of the Results **10**
- 2.1 Evaluation . . . 10
- 2.2 Comparison and Assessment of the Material Specimens . . . 12

3 Test Protocol **14**

A Data Management Plan **15**

Note on the use of tools:

The language model Claude (Anthropic) was used as a tool for drafts and evaluation scripts. The structure and tasks of the report follow the assignment of the chair. The report was written in LaTeX. Test execution, specimen dimensions, measurement data and photos are the group's own. All contents, calculations and results were checked and revised by the group, which takes responsibility for them.

Note on author contributions:

Georg Müller	Main author of Sections 1 and 2 (Description of the test objects, Interpretation of the results), plotting of the diagrams, formatting in LaTeX, preparation of the DMP using the TUM template
Junjie Qiu	Review of the diagrams in Section 1.6 (Plotting) and revision of the CFRP 0° diagrams; support in revising Sections 2.1 (a–b) and 2.2 (a, b, d, e) and adjustment of the corresponding entries in the test protocol
Mark Radlach	Main author of Section 3 (Test protocol); support for Sections 1.3 (Sources of error), 1.4 (Force measurement) and 1.5 (Strain measurement methods – transducers)
Martin Streicher	Review of the calculated results and proofreading

1 Description of the Test Objects

1.1 Working Principle of the Testing Machine

The tests were carried out on an electromechanical universal testing machine, type Instron 4505 (rated force 100 kN). Figure 1 shows the basic setup. The load frame consists of a rigid base, two columns, each with a lead screw, and a fixed upper crosshead. An electric motor in the base drives both screws synchronously via a gearbox and toothed belt. The screw nuts sit in the moving crosshead (lower cross member), which therefore moves vertically.

The specimen is clamped between two grips. The upper grip hangs from the fixed upper crosshead via the load cell, while the lower grip sits on the moving crosshead. When the crosshead moves down, the specimen is stretched and the tensile force is measured by the load cell. Since the load cell is mounted on the fixed crosshead, it does not move; cables and measuring chain stay at rest. The crosshead travel (crosshead displacement) is recorded by a rotary encoder on the drive. In addition, the strain can be measured locally on the specimen with an extensometer or strain gauges. Control and data acquisition run via the machine electronics and a connected PC; in the test, all channels were recorded at 100 Hz.

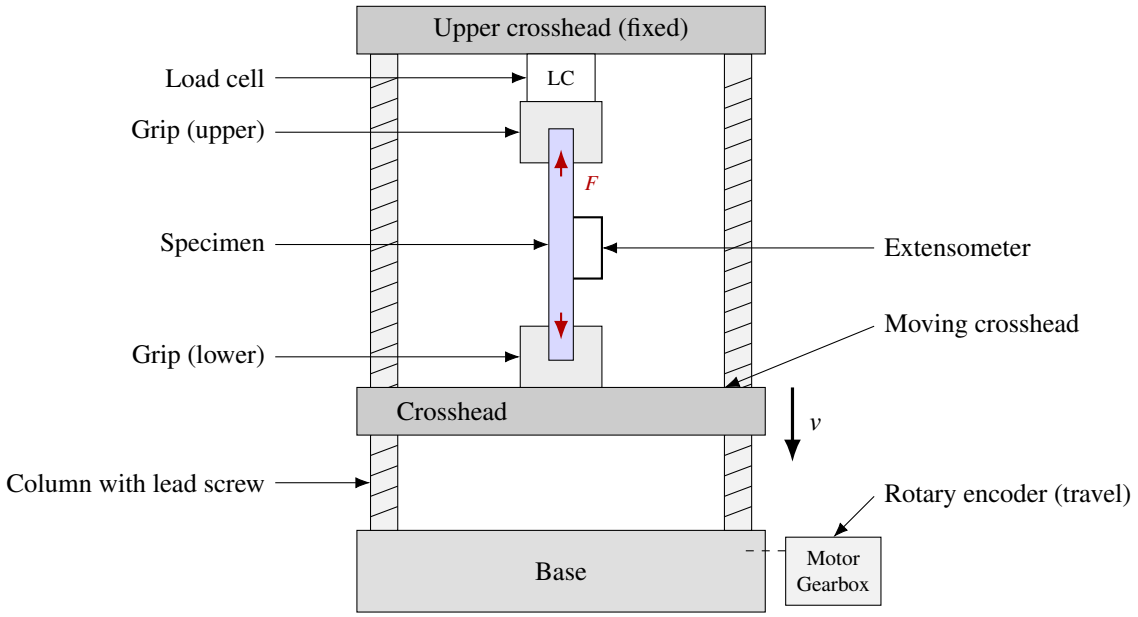


Figure 1: Schematic of the electromechanical tensile testing machine (Instron 4505). The lower crosshead is moved down by two lead screws; the load cell (LC) is mounted on the fixed upper crosshead. The tensile force F flows from the crosshead through the grips, specimen and LC into the upper crosshead and back into the base via the columns.

1.2 Control, Drive and Clamping

Control modes. The drive can be controlled by three quantities:

- **Displacement control:** The crosshead moves at constant speed. This is the standard case and was also used in the test. The measurement data give a testing speed of about 0.96 mm/min for the elastic aluminium measurement and the 90° specimen, and about 5.04 mm/min for all other tests to fracture.
- **Force control:** The force is increased at a constant rate or held at a value (e. g. for creep or hold phases). After the maximum force is reached, this control becomes unstable because the machine cannot follow the decreasing force.
- **Strain control:** The controller uses the extensometer signal. This sets the strain rate in the specimen directly, independent of the compliance of the machine. Some standards for metallic materials require this in the range of the proof stress.

Drive. A speed-controlled electric motor drives the two lead screws via a gearbox and belt, converting the rotation into a linear motion of the crosshead. Electromechanical drives are very well suited for quasi-static tests: the speed can be set finely and reproducibly, and there is no hydraulic system to maintain. For high frequencies (fatigue tests), servo-hydraulic machines are the better choice.

Clamping (wedge grip principle). The specimen is held in mechanical wedge grips (Fig. 2). Two wedges with serrated clamping faces sit in a housing with an inclined guide. They are first preloaded by hand. Under the tensile force F, the specimen pulls the wedges deeper into the housing; the incline (wedge angle α) creates a normal force N on the specimen that grows with the tensile force. Neglecting friction between wedge and housing, approximately

$$N \approx \frac{F}{2 \tan \alpha}. \tag{1}$$

The clamping is therefore self-reinforcing: the higher the load, the tighter the grip, as long as the static friction between jaw and specimen ($2\mu N \geq F$) is not exceeded. With CFRP specimens, the serrations of the jaws would damage the fibres and cause fracture in the grip. The CFRP specimens therefore have bonded GFRP end tabs that introduce the clamping force into the specimen over a larger area.

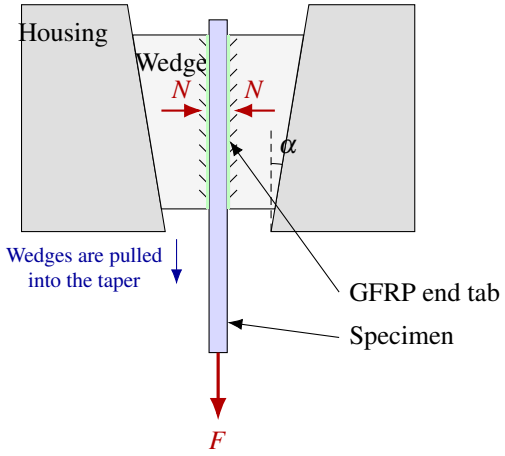


Figure 2: Wedge grip of the upper grip (section; the lower one is mirrored). The tensile force F pulls the wedges into the conical guide, so that the normal force N on the specimen also increases with the load.

1.3 Sources of Error

(a) **Possible errors during the test or caused by the testing machine**

- *Machine compliance:* The load frame, screws, load cell and grips also deform elastically. The crosshead displacement contains this deformation and clearly overestimates the specimen strain.
- *Seating and slipping in the grips:* The wedges settle at the start of the test and the serrations press into the specimen or end tab. This appears as a non-linear initial region of the force–displacement curve (e. g. Fig. 4).
- *Deformation outside the gauge length:* Specimen, end tabs and adhesive layer between the grips also stretch; the reference length for the crosshead displacement is therefore not clearly defined.
- *Misalignment:* If the grips are not exactly coaxial or the specimen is clamped at an angle, bending stresses occur in addition to the intended uniaxial loading. These distort the strain measurement and can cause premature fracture.
- *Failure of the load introduction:* Fracture in the grip or debonding of the end tabs. Both occurred in Group 1 (90° and 0°/90° specimen, see Section 1.6).

- *Sensor errors:* misaligned or poorly bonded strain gauges, temperature drift, slipping of the extensometer knife edges, zero offsets of the channels.
- *Geometry:* Measurement inaccuracy in width and thickness enters stress and Young's modulus directly; thickness variations are particularly relevant for CFRP.
- *Environment and testing speed:* Temperature and humidity mainly affect the matrix; an excessive testing speed changes the characteristic values.

(b) **Remedies**

- Measure strain locally on the specimen (extensometer, strain gauges) instead of via the crosshead displacement, or subtract the machine compliance using a calibration test.
- Determine Young's modulus between two limits rather than from zero (0.01 %/0.1 % strain for aluminium, 10 %/50 % $F_{\max}$ for CFRP). This removes seating and zero-point errors.
- Apply a small preload before the measurement starts.
- Align the grips and clamp the specimen centrally; if needed, apply strain gauges on front and back to check for bending.
- Bond end tabs carefully (surface preparation, full curing), use chamfered tabs to reduce stress peaks; discard specimens that break in the grip.
- Measure specimen dimensions at several points with a micrometer and average; test at least five specimens per variant.
- Calibrate the load cell and maintain the test equipment regularly.
- Keep to and document the standard atmosphere and the prescribed testing speed.
- When clamping the AlMg3 specimen, use a prefabricated jig that allows nearly uniaxial positioning centred in the jaws. Alternatively, and for the CFRP specimens, calculate the central position from the jaw and specimen geometry at the top and bottom and measure it with a caliper.

1.4 Force Measurement

The force is measured with a strain-gauge-based load cell. It consists of a metallic spring element that sits in the force flow between the fixed crosshead and the upper grip and deforms elastically and in proportion to the force under load. Strain gauges are applied to the spring element and wired as a Wheatstone full bridge: two gauges are stretched, two compressed (or stretched transversely). The relative change in resistance $\Delta R/R = k\,\varepsilon$ ($k \approx 2$) leads to a bridge output

$$\frac{U_A}{U_B} = \frac{k}{4}\,(\varepsilon_1 - \varepsilon_2 + \varepsilon_3 - \varepsilon_4), \tag{2}$$

which is amplified and digitised. Compared with a single gauge, the full bridge doubles or quadruples the signal and compensates for temperature effects as well as bending and transverse forces. The load cell is calibrated against reference weights or a standard; the output voltage is then directly proportional to the force.

1.5 Strain Measurement Methods – Transducers

Three methods were used in the test. Table 1 summarises them.

Strain gauges (SG). A meander-shaped metal foil grid on a carrier film is bonded to the specimen and stretches with it. Its electrical resistance changes in proportion to the strain ($\Delta R/R = k\,\varepsilon$). Since the relative change in resistance is very small, only the resulting voltage change is recorded using a Wheatstone bridge circuit. Strain gauges were used on the CFRP 0° and CFRP 0°/90° specimens.

Extensometer. Two knife edges are clamped to the specimen at the initial gauge length l_0 (12.5, 25 or 50 mm). If their distance changes, a flexible beam in the housing deforms, whose strain is in turn measured with strain gauges. Measuring range ± 5 mm, resolution about 10^{-5}. Because the extensometer is fragile and expensive, it was only used in tests with clearly lower fracture energy: in Test 1 on the AlMg3 specimen to determine Young's modulus, and in Test 4 on the CFRP UD 90° specimen to fracture.

Crosshead displacement. The crosshead travel is recorded without an additional sensor and related to a reference length. It contains all deformations in the force flow and is therefore only suitable as a rough approximation.

Table 1: Comparison of the strain transducers (costs as order of magnitude).

	Strain gauge	**Extensometer**	**Crosshead displacement**
Principle	Change in resistance of a bonded measuring grid	Change in distance between two knife edges, measured via strain gauges on a flexure beam	Rotary encoder on the drive, crosshead travel
Cost	a few to several tens of € per gauge, plus adhesive, wiring and application time; single use	several thousand €, but reusable	no additional cost
Advantages	very accurate, local, measures up to fracture, even at high fracture energy	quick to attach, reusable, averages over l_0	always available, unlimited measuring range
Disadvantages	time-consuming application, local only, not reusable, limited measuring range	can be damaged by high-energy fracture, knife edges can slip	includes machine and grip deformation, large errors

1.6 Plotting

The measurement data were imported, evaluated and plotted with the MATLAB script `evaluation.m` (based on the import script provided by the chair). The evaluation points from Section 2.1 are marked accordingly. For CFRP 0°, the red dot marks the maximum force; in the other fracture diagrams it marks the selected fracture point. The limits of the Young's modulus determination are shown as green dashed lines. The CFRP 0° diagrams also include the data after the maximum force; this time interval is shown as a grey dashed line.

Particularities of the Group 1 measurements. The CFRP 0°/90° specimen of Group 1 could not be tested to fracture because the bond of a GFRP end tab failed prematurely and the tab slipped off (Fig. 14). The force only rose to about 13.22 kN and then decreased gradually as the tab debonded. As instructed, the evaluation uses the measurement data of the 0°/90° specimen of Group 3 (fracture at 32.86 kN); the Group 1 curve is shown in grey in Fig. 10 and 11 for documentation.

The 90° specimen of Group 1 broke in the grip area at only 0.54 kN (Group 3: 1.85 kN). A fracture outside the gauge length does not reflect the material behaviour; the strength and elongation at break of this specimen are therefore of limited validity. The Group 3 curve is shown in grey for comparison.

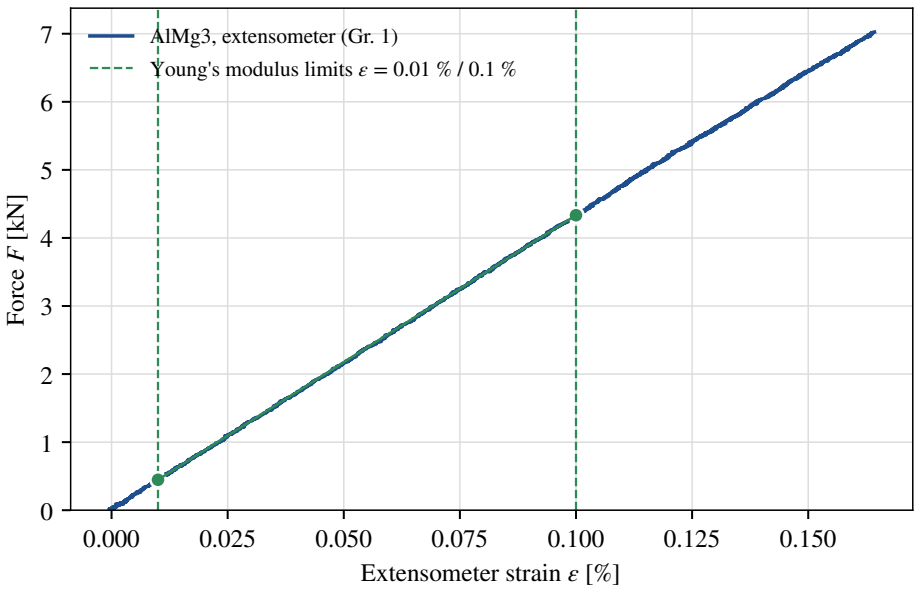


Figure 3: (a) AlMg3 elastic: force vs. extensometer strain. Limits of the Young's modulus determination at $\varepsilon = 0.01\,\%$ and $0.1\,\%$.

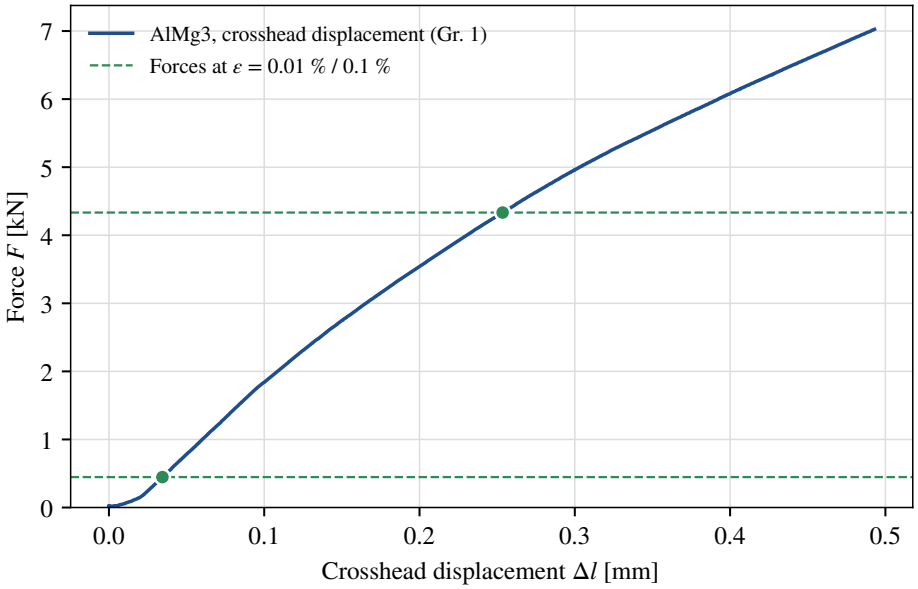


Figure 4: (b) AlMg3 elastic: force vs. crosshead displacement. The forces at $\varepsilon = 0.01\,\%$ and $0.1\,\%$ (extensometer) are transferred as limits.

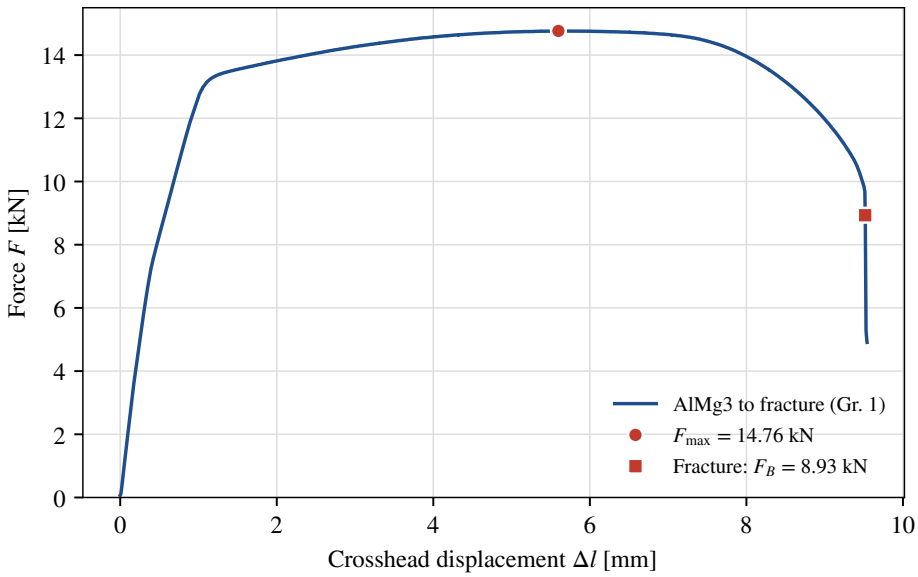


Figure 5: (c) AlMg3 fracture: force vs. crosshead displacement with maximum force F_{max} and fracture point.

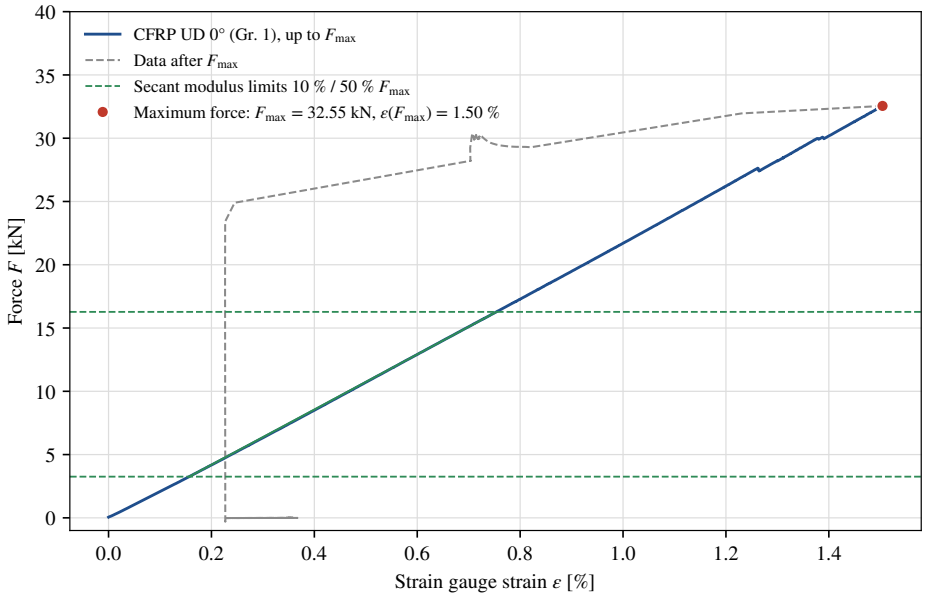


Figure 6: (d) CFRP 0°: force vs. strain gauge strain with the complete measurement. Blue: up to the maximum force; grey dashed: afterwards. The red dot marks the maximum force.

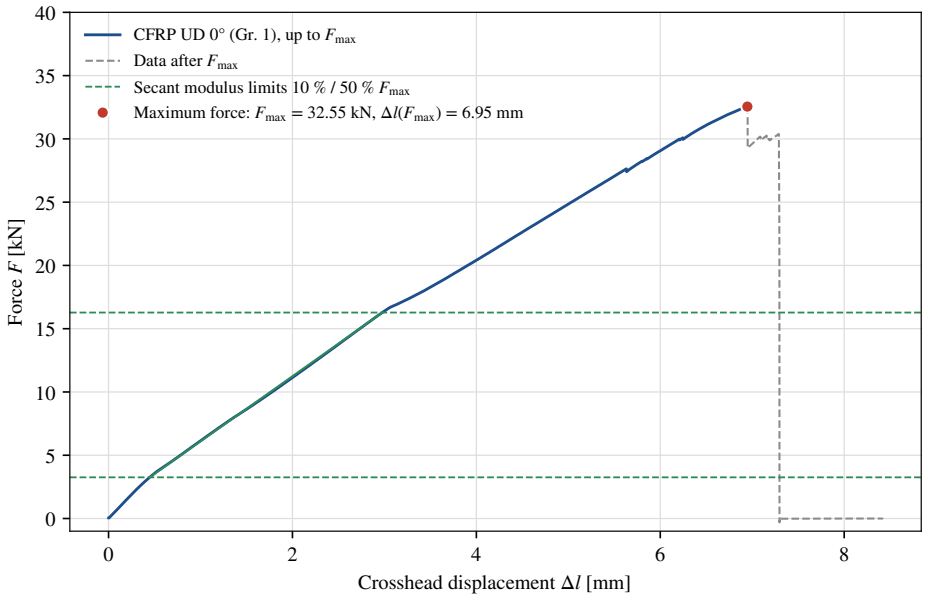


Figure 7: (e) CFRP 0°: force vs. crosshead displacement with the complete measurement. Blue: up to the maximum force; grey dashed: afterwards. The red dot marks the maximum force.

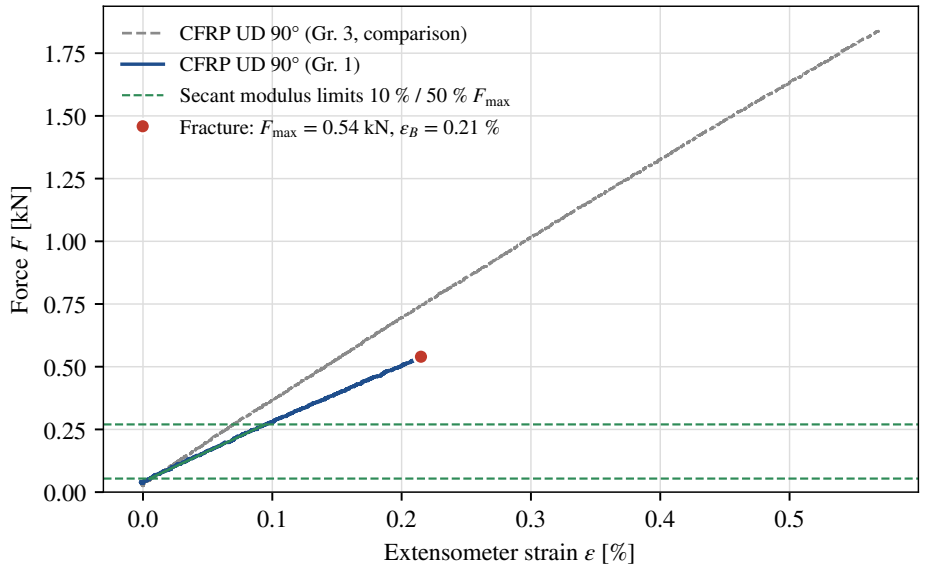


Figure 8: (f) CFRP 90°: force vs. extensometer strain (Group 1, fracture in the grip; Group 3 for comparison).

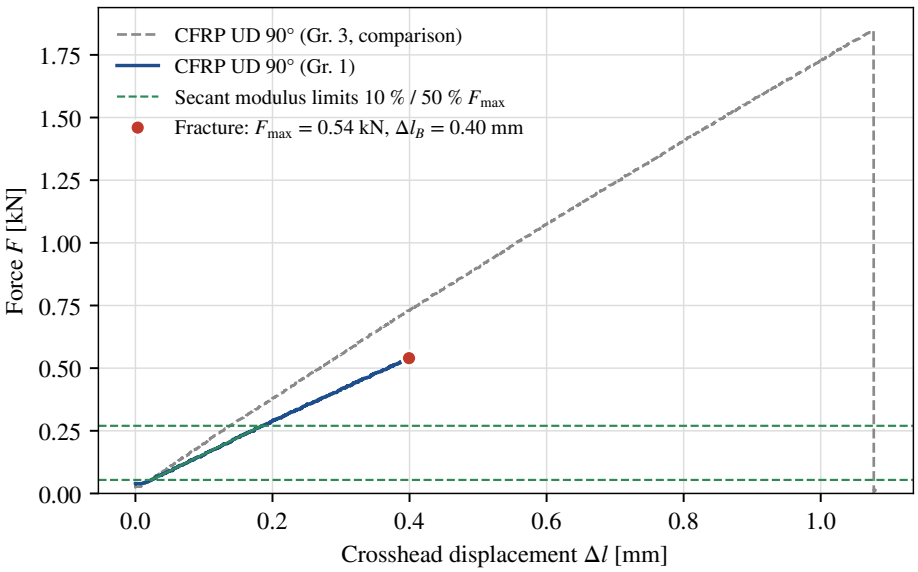


Figure 9: (g) CFRP 90°: force vs. crosshead displacement.

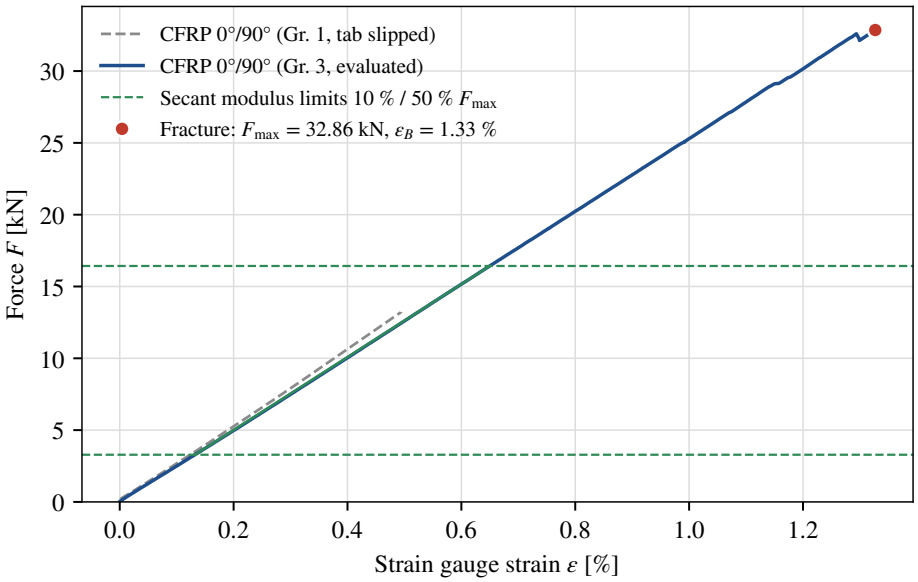


Figure 10: (h) CFRP 0°/90°: force vs. strain gauge strain (evaluated: Group 3; Group 1 with end tab failure in grey).

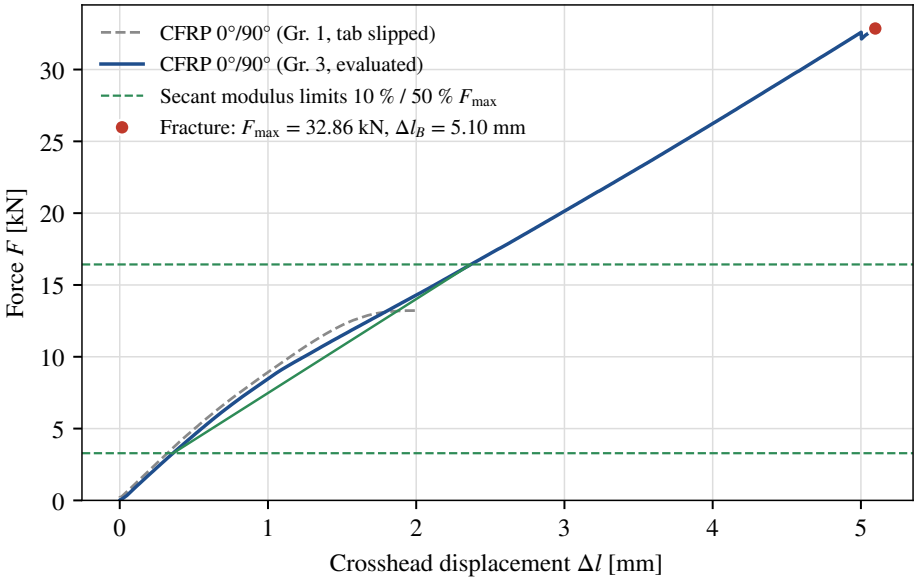


Figure 11: (i) CFRP 0°/90°: force vs. crosshead displacement.

2 Interpretation of the Results

2.1 Evaluation

The evaluation is based on the specimen dimensions from the test protocol (Table 5). For the 0°/90° specimen, the measurement data of Group 3 are used together with the dimensions of the identical specimen from our own protocol; differences in cross-section between the two specimens therefore enter stress and Young's modulus directly. The initial cross-section is $A_0 = b \cdot t$; the strain from the crosshead displacement is calculated as $\varepsilon_{CH} = \Delta l / L_0$ with the reference length L_0.

(a) **Fracture stress.** The fracture stress follows from the fracture force and the initial cross-section, $\sigma_B = F_B / A_0$. For the CFRP 0° specimen, the maximum nominal stress is determined from the maximum force. After $F_{max} = 32.55\,\text{kN}$ is reached, a considerable load-bearing capacity initially remains. The maximum force point is therefore distinguished from the later complete force drop. For AlMg3, the tensile strength $R_m = F_{max}/A_0$ and the stress at the fracture point F_B/A_0 are given separately, since the force decreases after necking until fracture. The fracture points are marked in Fig. 5, 8 and 10; for CFRP 0°, the maximum force point is marked in Fig. 6 and 7.

Table 2: Fracture forces and fracture stresses.

Specimen	F_{max} [kN]	F_B [kN]	A_0 [mm^2]	σ_B [N/mm^2]
AlMg3	14.76	8.93	64.94	R_m = 227; 138
CFRP UD 0°	32.55	32.55	14.20	2292
CFRP UD 90°*	0.54	0.54	31.15	17
CFRP 0°/90°**	32.86	32.86	31.81	1033

* Fracture in the grip, limited validity. ** Measurement data of Group 3.

CFRP 0°: The force, stress, strain and displacement values refer to the maximum force point. Fracture force and elongation at break were not determined independently of it.

(b) **Elongation at break of the CFRP specimens.** The strain and displacement values are read at the marked evaluation point from the strain gauge or extensometer and the crosshead displacement (Fig. 6–11). For CFRP 0°, the values refer to the maximum force point: $\varepsilon(F_{max}) = 1.504\,\%$ and

$\Delta l(F_{\max}) = 6.95$ mm. No independently determined elongation at break is given for this specimen. The error factor is $f_\varepsilon = \varepsilon_{\mathrm{SG/Ext}}/\varepsilon_{\mathrm{CH}}$.

Table 3: Elongation at break of the CFRP specimens and error factor.

Specimen	$\varepsilon_{B,\mathrm{SG/Ext}}$ [%]	Δl_B [mm]	L_0 [mm]	$\varepsilon_{B,\mathrm{CH}}$ [%]	f_ε [–]
CFRP UD 0°	1.504	6.95	150.6	4.614	0.33
CFRP UD 90°*	0.215	0.40	151.6	0.263	0.82
CFRP 0°/90°**	1.327	5.10	154.6	3.297	0.40

CFRP 0°: The force, stress, strain and displacement values refer to the maximum force point. Fracture force and elongation at break were not determined independently of it.

The error factor is smaller than one, so the strain from the crosshead displacement is larger than the actual specimen strain. Besides the elongation of the specimen within the gauge length, the crosshead displacement also contains the elastic deformation of the load frame, screws, load cell and grips, the seating and slipping of the wedges, the deformation of the end tabs and adhesive layers, and the strain of the specimen outside the gauge length. All of these contributions are wrongly attributed to the reference length. Strain gauges and extensometer, in contrast, measure directly on the specimen.

(c) **Young's modulus.** For AlMg3, Young's modulus is determined between the strain limits $\varepsilon_{0.01} = 0.01\,\%$ and $\varepsilon_{0.1} = 0.1\,\%$ (Fig. 3, 4):

$$E = \frac{F_{\varepsilon 0.1} - F_{\varepsilon 0.01}}{A_0\,(\varepsilon_{0.1} - \varepsilon_{0.01})} = \frac{F_{\varepsilon 0.1} - F_{\varepsilon 0.01}}{A_0 \cdot 0.09\,\%}, \tag{3}$$

with $F_{\varepsilon 0.01} = 0.447$ kN and $F_{\varepsilon 0.1} = 4.333$ kN. For the CFRP specimens, the secant modulus between 10 % and 50 % of the fracture force is determined according to EN 2561 (Fig. 6–11):

$$E = \frac{F_{50} - F_{10}}{A_0\,(\varepsilon_{F50} - \varepsilon_{F10})} = \frac{0.4\,F_{\max}}{A_0\,(\varepsilon_{F50} - \varepsilon_{F10})}. \tag{4}$$

For the crosshead displacement, the same force limits are used and the strains are replaced by $\Delta l/L_0$. The error factor is $f_E = E_{\mathrm{SG/Ext}}/E_{\mathrm{CH}}$.

Table 4: Young's moduli and error factor.

Specimen	Limits	$E_{\mathrm{SG/Ext}}$ [GPa]	E_{CH} [GPa]	f_E [–]
AlMg3	0.01–0.1 % ε	66.5	17.8	3.74
CFRP UD 0°	3.26–16.28 kN	153.5	54.6	2.81
CFRP UD 90°*	0.05–0.27 kN	7.7	6.5	1.19
CFRP 0°/90°**	3.29–16.43 kN	79.8	31.8	2.51

The error factor f_E is larger than one: the modulus from the crosshead displacement is too low because the denominator contains the same excessive strain as in (b). Since only a force interval is considered here, zero-point and seating errors are largely eliminated; the compliance of the machine, grips and end tabs, however, remains fully included. The stiffer the specimen relative to the machine, the larger the error. The values confirm this: f_E is largest for the thick aluminium specimen (3.74), followed by CFRP 0° (2.81) and 0°/90° (2.51); for the very compliant 90° specimen it is smallest at 1.19.

2.2 Comparison and Assessment of the Material Specimens

The reference values are the manufacturer data of the prepreg system KUBD 1409 (UTS fibre, epoxy resin, fibre volume fraction approx. 60 %): $E_0 \approx 148$ GPa, $R_0 \approx 2100$ N/mm^2, $\varepsilon_{B,0} \approx 1$ %; $E_{90} \approx 9$ GPa, $R_{90} \approx 34$ N/mm^2, $\varepsilon_{B,90} \approx 0.4$–$0.5$ %.

(a) **Young's modulus of the CFRP specimens.** The measured values are $E_0 = 153.5$ GPa, $E_{0/90} = 79.8$ GPa and $E_{90} = 7.7$ GPa. The order $E_0 > E_{0/90} > E_{90}$ follows from the load distribution. In the 0° specimen, all fibres are aligned with the load and the stiffness is fibre-dominated. In the 90° specimen, the fibres are transverse and the load has to be carried by the matrix and the fibre–matrix interface; the stiffness is matrix-dominated and about one order of magnitude lower. The laminate $[0/90/0/90/0]_s$ contains six 0° and four 90° plies. A simple rule of mixtures over the ply fractions gives an expected value of about $0.6 \cdot 148 + 0.4 \cdot 9 \approx 92$ GPa. Here too, the 0° fibres dominate. E_0 agrees well with the manufacturer value, E_{90} is slightly lower. The somewhat lower value of the laminate could be related, among other things, to a different fibre volume fraction. A higher resin content lowers the fibre volume fraction and thus the stiffness, all else being equal. Whether this is actually the cause for the tested specimen cannot be concluded from the greater thickness alone. In addition, the measurement data and specimen dimensions come from different specimens.

(b) **Elongation at break of the CFRP specimens.** The measured values are $\varepsilon_0(F_{\max}) = 1.504$ %, $\varepsilon_{B,0/90} = 1.327$ % and $\varepsilon_{B,90} = 0.215$ % (Group 3: 0.571 %). For CFRP 0°, a strain of 1.504 % was measured at the maximum force point. Since a considerable load-bearing capacity remains afterwards, this value is not simply equated with the fibre elongation at break. At 90°, the matrix or the fibre–matrix interface fails at low strain (matrix-dominated). Because of the fracture in the grip, the Group 1 value is clearly below the expected range, while the Group 3 value is within it. In the 0°/90° laminate, the 90° plies crack first (inter-fibre fracture at about the elongation at break of the 90° specimen), but the laminate continues to carry load via the 0° plies. The elongation at break of the laminate is therefore determined by the fibres and is of the same order as that of the 0° specimen.

(c) **Young's modulus of AlMg3 compared with CFRP.** For AlMg3, $E = 66.5$ GPa (literature approx. 70 GPa). As expected, the value is clearly below E_0 and $E_{0/90}$, but far above E_{90}. Aluminium is isotropic: its stiffness is the same in all directions, whereas CFRP is stiffer only in the fibre direction. Relative to density ($\rho_{\mathrm{Al}} \approx 2.7$ g/cm^3, $\rho_{\mathrm{CFRP}} \approx 1.5$–$1.6$ g/cm^3), the advantage of CFRP in the fibre direction is even greater. *Conclusion:* CFRP makes sense when the main load directions are known and the fibres can be aligned accordingly, e. g. in tension struts, spars or pressure vessels. For multiaxial, changing or unknown load directions, at load introduction points, and where damage tolerance and reparability are important, aluminium is often the simpler and more robust choice. A quasi-isotropic CFRP laminate has a stiffness of the same order as aluminium but is considerably lighter.

(d) **Tensile strength of AlMg3 and CFRP.** The measured values are $R_{m,\mathrm{Al}} = 227$ N/mm^2, $\sigma_{B,0} = 2292$ N/mm^2, $\sigma_{B,0/90} = 1033$ N/mm^2 and $\sigma_{B,90} = 17$ N/mm^2. $R_{m,\mathrm{Al}}$ matches the literature value of about 225 N/mm^2; $\sigma_{B,0}$ is slightly above the manufacturer value of 2100 N/mm^2. Because of the fracture in the grip, $\sigma_{B,90}$ is only about half the manufacturer value (34 N/mm^2) and is not representative. Fibre orientation has a very large influence in CFRP: in the fibre direction, the strength exceeds that of AlMg3 many times over, while transverse to it, it is far lower because failure is mainly governed by the matrix and the fibre–matrix interface. The 0°/90° laminate lies in between, roughly according to the fraction of 0° plies. A quasi-isotropic laminate (e. g. $[0/\pm 45/90]_s$) has more direction-independent properties and reaches about the strength of high-strength aluminium alloys at a lower density. Unlike aluminium, however, CFRP fails in a brittle manner without plastic reserve and is sensitive to impact and notches. Aluminium yields before fracture and reduces stress peaks through plastic flow.

(e) **Fracture patterns and failure modes.**

- *AlMg3 (Fig. 12):* ductile failure. The curve (Fig. 5) shows a linear elastic region, yielding with strain hardening up to F_{max}, then necking with decreasing force and finally fracture. The photos show the clear local necking in width and thickness and a fracture surface inclined to the load direction (shear fracture in the necking zone). In the clamping area, the marks of the serrated jaws are visible; the specimen is undeformed there.
- *CFRP 0° (Fig. 13):* brittle, fibre-dominated failure. The curve is almost linear up to the maximum force point (Fig. 6). Small force drops are visible; their exact cause cannot be determined from the measurement signals alone. After the maximum force, a considerable load-bearing capacity initially remains before the force drops to almost zero. Fracture occurred suddenly with a high release of energy: the specimen split longitudinally into individual fibre bundles and strips over its entire free length ("broom-like fracture"). The splits are caused by matrix cracks parallel to the fibres, which dissipate the released energy. The end tabs remained intact.
- *CFRP 90°:* brittle, matrix-dominated failure at low force and without warning; the curve (Fig. 8) is almost linear up to fracture. Such an inter-fibre fracture typically runs smoothly and perpendicular to the load direction between the fibres. In Group 1, the fracture occurred in the grip; no photo of the fracture pattern is available.
- *CFRP 0°/90° (Fig. 14):* In Group 1, the specimen did not break; instead, a GFRP end tab slipped off. The bond between tab and specimen failed; adhesive residue is visible on the specimen surface and on the edge of the tab, while the specimen itself is undamaged. In the diagram, this shows as a gradual force drop after about 13.22 kN instead of a sudden fracture (Fig. 10). With regular failure (Group 3), the curve is almost linear up to fracture; the inter-fibre fractures of the 90° plies reduce the stiffness only slightly, as the 0° plies carry the load. The final failure is a brittle fibre fracture with delaminations between the plies.



Figure 12: AlMg3 after fracture: left and centre still clamped, with necking and inclined fracture surface; right, one specimen half with jaw marks in the clamping area.

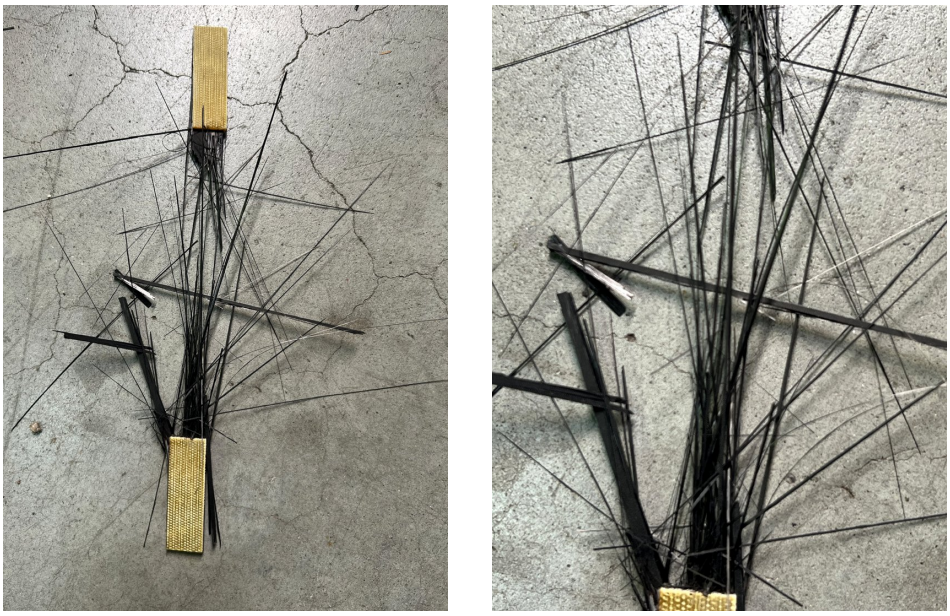


Figure 13: CFRP UD 0° after fracture: specimen split into fibre bundles over its entire free length (“broom-like fracture”), detail on the right.

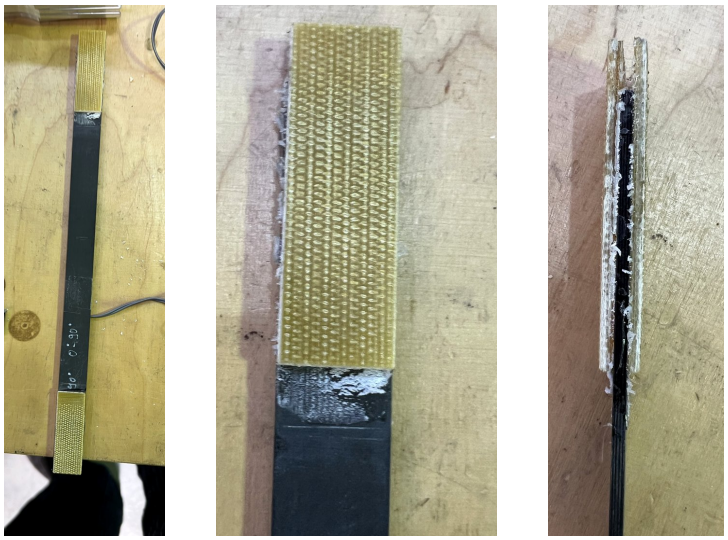


Figure 14: CFRP 0°/90° of Group 1: left, the unbroken specimen with strain gauge; centre and right, the slipped GFRP end tab with adhesive residue (front and side view).

3 Test Protocol

Table 5: Specimen dimensions and sensors. Thickness measured top/centre/bottom, $\bar{t}$ mean value.

No.	Material	b [mm]	Thickness [mm]	$\bar{t}$ [mm]	L_0 [mm]	Sensors
1	AlMg3	16.06	4.04 / 4.04 / 4.05	4.043	65.0	Extensometer, crosshead
3	CFRP UD 0	16.20	0.88 / 0.87 / 0.88	0.877	150.6	SG, crosshead
4	CFRP UD 90	16.25	1.95 / 1.90 / 1.90	1.917	151.6	Extensometer, crosshead
5	CFRP 0/90**	15.85	2.00 / 2.00 / 2.02	2.007	154.6	SG, crosshead

Table 6: Notes, events and fracture pattern.

No.	Notes / events / fracture pattern / other
1	AlMg3 specimen purely elastic with calibrated extensometer up to approx. 7 kN (approx. 0.96 mm/min).
2	Load on AlMg3 specimen reduced to 60 N in manual mode. Extensometer removed. Crosshead displacement zeroed again. Displacement-controlled at constant speed to fracture (approx. 5.04 mm/min). $F_{max} = 14.76$ kN. Fracture pattern as expected: necking and inclined fracture surface (Fig. 12).
3	CFRP UD 0° specimen with strain gauge, displacement-controlled to fracture (approx. 5.04 mm/min). Maximum force of 32.55 kN at 101.34 s. A considerable load-bearing capacity then remains for about four seconds; between 105.52 s and 105.53 s the force drops to almost zero. Fracture pattern: specimen heavily split in the longitudinal direction. GFRP end tabs intact (Fig. 13).
4	CFRP UD 90° specimen with extensometer, displacement-controlled to fracture (approx. 0.96 mm/min). Fracture in the grip area at 0.54 kN; approx. 1 kN was expected. The test is therefore of limited validity and would have to be repeated. Due to time constraints, we continued with the next test. No photo of the fracture pattern available.
5	CFRP 0/90 specimen with strain gauge to fracture (approx. 5 mm/min). Group 1 had two faulty test runs. First, the specimen was not clamped firmly enough, so the force decreased during the test while the crosshead displacement increased. The specimen was therefore unloaded in manual mode, preloaded again and the crosshead displacement zeroed again. The test was then restarted at the same speed. The same faulty behaviour was observed, after which the test was stopped and the specimen inspected: GFRP end tab slipped at 13.22 kN (bond failed, Fig. 14), no fracture of the specimen. Evaluation with the data of Group 3 (fracture at 32.86 kN).

A Data Management Plan

DMP submitted by

Group 1: Georg Müller, Junjie Qiu, Mark Radlach, Martin Streicher, Technical University of Munich

Submitted on

27.10.2026

Project name

Tensile test: test execution and research data management

Formal context of the project

This project is part of the lab course *Scientific Experiment with Research Data Management – Tensile Test* (Wissenschaftlicher Versuch mit Forschungsdatenmanagement – Zugversuch), offered by the Chair of Product Development and Lightweight Design (LPL), TUM School of Engineering and Design, Technical University of Munich, in the winter semester 2026/27. It is a teaching project within the B.Sc. Mechanical Engineering programme and is not embedded in a larger research collaboration. No external partners are involved and the project receives no external funding, so a funding agency and project ID do not apply. As instructed by the course organisers, the dataset of Group 1 is complemented by the data file of the CFRP 0°/90° specimen of Group 3.

Mandated policies (TUM only)

- `https://www.it.tum.de/en/projects/research-data-management/`
- `https://www.ub.tum.de/en/open-access-policy`
- `https://www.forte.tum.de/fileadmin/w00bgt/www/Patent-_und_Lizenzbuero/tum_patentpolitik_ip_policy_en.pdf`

Principal investigators

Prof. Dr. Markus Zimmermann, LPL, Technical University of Munich

Further researchers

Georg Müller, Junjie Qiu, Mark Radlach, Martin Streicher, all Technical University of Munich

Begin and end of project

12.10.2026 – 27.10.2026

Project description

Tensile tests on AlMg3 and CFRP (UD 0°, UD 90°, $[0/90/0/90/0]_s$) on an Instron 4505 to determine fracture stress, elongation at break and Young's modulus, to compare isotropic and anisotropic materials and to quantify the error of strain measurement via crosshead travel compared with extensometer and strain gauges. Details are given in Sections 1 and 2 of this report.

Relevant research data

Raw data (force, crosshead displacement, extensometer or strain gauge signal, time stamps) of five tests of Group 1 and the CFRP 0°/90° test of Group 3; test protocol with specimen dimensions; evaluation script; figures; characteristic values; report.

Reuse of existing data

Raw data of Group 3 and the MATLAB import script, provided by the LPL via Moodle; manufacturer data of the prepreg system KUBD 1409 from the course script.

Data format

Raw data: MATLAB MAT files (version 5), additionally exported to CSV for archiving. Results: CSV (`results.csv`) and LaTeX macros (`results.tex`). Figures: PDF. Report: LaTeX source and PDF.

Software for data processing

MATLAB (R2020a or newer) with the evaluation script `evaluation.m`, based on the import script of the LPL; LaTeX for the report.

Size of data

Ten MAT files with about 5 MB in total; the complete project stays below 50 MB.

Storage and backup

Originals remain on Moodle; working copy in a shared TUM Sync+Share folder (LRZ) with local copies of all group members.

Documentation and metadata

Test protocol, `README` with folder structure, variables and units and comments in the evaluation script. Recorded metadata per test: date, group, material, layup, specimen dimensions, reference length, sensors, sampling rate (100 Hz), testing speed, machine, standard and validity.

Reproducibility and quality assessment

The evaluation is performed using `evaluation.m`, which generates all characteristic values and figures from the raw data, including the two CFRP-0° diagrams with the measurements after the maximum force. The results and figures can be reproduced with this script and the input data. Quality checks: comparison with manufacturer and literature values, comparison with Group 3, check of the testing speed and inspection of the fracture location. Invalid tests (tab failure of the Group 1 0°/90° specimen) and tests of limited validity (fracture of the Group 1 90° specimen in the grip) are flagged.

File naming, organisation and versioning

Folders `daten/` (raw data per group, original names), `figures/` and `photos/`; evaluation script and results in the project root. Raw data are never modified; script and report are versioned with git.

Data sharing, sensitive data, archiving, publication

All group members have access to the full dataset. The data contain no personal or confidential information. Raw data, script, results and report are kept for ten years according to the TUM rules of good scientific practice. The data are not published; they are submitted with the report.

Responsibility and costs

Overall responsibility: principal investigator; data collection: all group members. No additional costs arise.

Last update of this DMP

27.10.2026